

COURSE PLAN

PYTHON PROGRAMMING

Open Elective

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UNIT-WISE COURSE PLAN

PYTHON PROGRAMMING

Course Type: Open Elective | **Total Hours:** 60 (15 weeks × 4 hours)

Software / Tools: Python 3, Jupyter Notebook / IDLE, pip, NumPy, Pandas, Matplotlib (Anaconda distribution)

Text Books: (1) Mark Summerfield, *Programming in Python 3: A Complete Introduction to the Python Language*, 2/e, Addison-Wesley; (2) Phuong Vo.T.H, Martin Czygan, *Python: End-to-End Data Analysis Learning Path, Module 1: Getting Started with Python Data Analysis*, Packt

UNIT I: Introduction to Python — 9 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Rapid Introduction to Procedural Programming – Python interpreter, IDLE / Jupyter, first programs	1	Lecture / Demonstration	Write and run simple Python programs
2	Data Types – Identifiers and Keywords, Integral Types, Floating-Point Types	1	Lecture / Board	Evaluate expressions with int, float, bool and complex values
3	Strings – Comparing Strings, Slicing and Striding Strings	1	Lecture / Worked Examples	Extract and reverse substrings using slices and strides
4	String Operators and Methods; String formatting with str.format	1	Lecture / Worked Examples	Format a report using str.format and f-strings
5	Collection Data Types – Tuples and Lists	2	Lecture / Demonstration	Build and manipulate lists and tuples of records
6	Sets and Dictionaries	1	Lecture / Worked Examples	Word-frequency count using dictionaries and sets
7	Iterating and Copying Collections – shallow vs. deep copy	1	Demonstration / Think-Pair-Share	Predict outputs of copy vs. reference programs
8	Introduction to PIP – installing and managing packages	1	Demonstration / Quiz	Install NumPy and Pandas with pip; quiz on Unit I

UNIT II: Control Structures and Functions — 8 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Python Control Structures – Conditional Branching	1	Lecture / Board	Programs on grading and leap-year checks
2	Looping – for, while, break, continue, else on loops; comprehensions	1	Lecture / Worked Examples	Generate number patterns and prime numbers
3	Exception Handling – try, except, else, finally; raising exceptions	1	Lecture / Worked Examples	Handle invalid input and file errors
4	Custom Functions – arguments, default and keyword arguments, *args/**kwargs, lambda, scope	2	Lecture / Problem Solving	Write reusable functions for common tasks
5	Python Library Modules – random, math, time, os, shutil, sys	1	Demonstration	Simulate dice rolls; list and copy files
6	Python Library Modules – glob, re, statistics	1	Demonstration / Worked Examples	Search files with glob; validate text with regular expressions
7	Creating a custom module	1	Demonstration / Quiz	Package own functions as a module; quiz on Unit II

UNIT III: Object Oriented Programming and File Handling — 8 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Object Oriented Concepts and Terminology	1	Lecture / PPT	Identify classes and objects in real-world scenarios
2	Custom Classes – Attributes and Methods; special methods	2	Lecture / Worked Examples	Design a Student / BankAccount class
3	Inheritance and Polymorphism	2	Lecture / Worked Examples	Build a Shape class hierarchy with overridden methods
4	Using Properties to Control Attribute Access	1	Lecture / Demonstration	Add validated properties to a class
5	File Handling – Writing and Reading Binary Data (pickle, struct)	1	Demonstration	Save and load objects with pickle
6	Writing and Parsing Text Files	1	Problem Solving / Quiz	Parse a text/CSV file into objects; quiz on Unit III

UNIT IV: NumPy Arrays and Vectorized Computation — 9 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	NumPy arrays – ndarray, data types; Array creation	1	Lecture / Demonstration	Create arrays using array, arange, zeros, linspace
2	Indexing and slicing; Fancy indexing	2	Lecture / Worked Examples	Select rows, columns and elements with boolean and integer arrays
3	Numerical operations on arrays; Array functions – universal functions, broadcasting	2	Lecture / Worked Examples	Vectorise loops; compare timing with plain Python
4	Data processing using arrays – conditional logic, statistics, sorting, unique	1	Demonstration / Problem Solving	Summarise a dataset with array methods
5	Loading and saving data – saving an array, loading an array (npz, npy, text)	1	Demonstration	Save and reload arrays in binary and text formats
6	Linear algebra with NumPy	1	Worked Examples	Matrix product, inverse, determinant, solving linear systems
7	NumPy random numbers	1	Demonstration / Quiz	Simulate random walks; quiz on Unit IV

UNIT V: Data Analysis with Pandas — 10 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	An overview of the Pandas package; Pandas data structures – Series	1	Lecture / Demonstration	Create and operate on Series objects
2	The DataFrame – creation, columns and rows	1	Lecture / Demonstration	Build DataFrames from dictionaries and CSV files
3	Essential Basic Functionality – Reindexing and altering labels, Head and tail, Binary operations	2	Lecture / Worked Examples	Reindex and align two DataFrames
4	Functional statistics; Function application; Sorting	1	Worked Examples	Apply functions and sort a dataset
5	Indexing and selecting data – loc, iloc, boolean selection	1	Lecture / Problem Solving	Query a dataset with conditions
6	Computational tools – covariance, correlation, ranking, rolling windows	1	Lecture / Worked Examples	Compute correlations and moving averages
7	Working with Missing Data – detecting, dropping and filling	1	Demonstration / Problem Solving	Clean a dataset with missing values
8	Advanced Uses of Pandas – Hierarchical indexing; the Panel data structure	2	Lecture / Quiz	Use MultiIndex for multi-level data; quiz on Unit V

UNIT VI: Data Analysis Application Examples — 6 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Data munging; Cleaning data	1	Case Study / Demonstration	Clean a raw real-world dataset
2	Filtering data	1	Worked Examples	Filter records by multiple conditions
3	Merging data – merge, join, concat	1	Worked Examples	Combine related tables
4	Reshaping data – pivot, melt, stack and unstack	1	Demonstration / Problem Solving	Convert between wide and long formats
5	Data aggregation and Grouping data – groupby, pivot tables	2	Case Study / Quiz	Produce group-wise summaries; quiz on Unit VI

UNIT VII: Data Visualization — 7 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	The matplotlib API primer – Line properties	1	Lecture / Demonstration	Plot functions with different line styles and colours
2	Figures and subplots	1	Demonstration	Create multi-panel figures
3	Exploring plot types – Scatter plots and Bar plots	1	Worked Examples	Visualise relationships and category comparisons
4	Histogram plots	1	Worked Examples	Plot distributions with different bin sizes
5	Legends and annotations	1	Demonstration / Think-Pair-Share	Annotate key points in a plot
6	Plotting functions with Pandas	2	Case Study / Quiz	Exploratory data analysis of a dataset; quiz on Unit VII

UNIT VIII: Revision and Assessment — 3 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Revision of Units I–IV – Python basics, functions, OOP and NumPy	1	Tutorial / Discussion	Solve previous question papers
2	Revision of Units V–VII – Pandas, data analysis and visualization	1	Tutorial / Discussion	Solve previous question papers
3	Assignment review, slip test and doubt clearance	1	Slip Test / Discussion	Written test and feedback on assignments

Overall Hour Distribution

Unit	Unit Title	Hours	Percentage
I	Introduction to Python	9	15.00%
II	Control Structures and Functions	8	13.33%
III	Object Oriented Programming and File Handling	8	13.33%
IV	NumPy Arrays and Vectorized Computation	9	15.00%
V	Data Analysis with Pandas	10	16.67%
VI	Data Analysis Application Examples	6	10.00%
VII	Data Visualization	7	11.67%
VIII	Revision and Assessment	3	5.00%
	GRAND TOTAL	60	100%

General Modes of Teaching

- Each session is a 1-hour lecture: concepts are introduced on the board or through PPT, followed by live coding in Jupyter Notebook.
- Every topic is illustrated with small programs that students run and modify during the class.
- Case studies on real datasets connect NumPy, Pandas and Matplotlib with practical data analysis.
- Active-learning techniques such as Think-Pair-Share and short quizzes check understanding at the end of each unit.
- Continuous assessment through programming assignments, quizzes, slip test and mid-semester examinations.
- Additional exercises are given to fast learners, and remedial sessions are held for slow learners.
- The final three hours are for revision, solving previous question papers and the slip test.